

Conjecture 00000007793 is false: the two-point distance variance ratio diverges

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1 The conjecture and its reading

Statement (English). *Definition: The variance of $R = |X - Y|$, the distance between two independent isotropic uniform ball points. Conjecture: $\text{Var}(R)/\text{Var}(|X|) = 2 - 2/\sqrt{3} + O(n^{-1})$ with first correction coefficient $-1/(3\sqrt{3})$; the ratio strictly improves the thin-shell constant in the two-body case, with equality only for X, Y iid uniform on the ball, any correlated coupling increasing the ratio.* The Chinese text says the same.

Reading. n is the dimension: it is the only parameter in the statement, and the expansion $O(n^{-1})$ concerns $n \rightarrow \infty$. X and Y are independent and both uniformly distributed on a ball of $\mathbb{R}^n$ centred at 0. Write

$$a_n = \frac{\text{Var } |X - Y|}{\text{Var } |X|}.$$

The first conjunct says $a_n - (2 - 2/\sqrt{3}) = O(1/n)$ as $n \rightarrow \infty$, i.e. there are C, N with $|a_n - (2 - 2/\sqrt{3})| \leq C/n$ for $n \geq N$. The statement is a conjunction, so refuting this first conjunct refutes it. We do not address the later conjuncts (the correction coefficient and the coupling clause) separately. Their natural readings presuppose the first conjunct.

Normalisation. “Isotropic” may mean the unit ball or the ball scaled to covariance I (radius $\sqrt{n+2}$). The ratio a_n is invariant under scaling $X \mapsto rX, Y \mapsto rY$ ($r > 0$), because both variances are multiplied by r^2 . We prove the refutation for an arbitrary radius sequence $r_n > 0$.

Conventions. $|\cdot|$ is the Euclidean norm. $\text{Var } Z = \mathbb{E}(Z - \mathbb{E}Z)^2$ (Mathlib’s `ProbabilityTheory.variance`). The uniform law on the open ball $B_r = \{x : |x| < r\}$ is Lebesgue measure restricted to B_r and normalised, $\text{volume}[\cdot \mid B_r]$ (Mathlib’s `ProbabilityTheory.cond`). Open and closed balls differ by a null set. Independence is Mathlib’s `IndepFun`. In the concrete model, (X, Y) is the pair of coordinates under the product measure.

2 Result

Theorem 1. *For X, Y independent and uniform on the unit ball of $\mathbb{R}^n$, $n \geq 1$,*

$$a_n \geq \frac{(n+1)^2}{4(n+2)} \geq \frac{n}{8}.$$

Hence $a_n \rightarrow +\infty$. In particular a_n has no finite limit, and $a_n - (2 - 2/\sqrt{3}) \neq O(1/n)$. The same holds for any radii $r_n > 0$.

Monte Carlo agrees: $a_1 = 8/3$ exactly, and a_n is about 7.4, 27 and 102 at $n = 10, 50, 200$, so roughly $n/2$. The claimed limit is $2 - 2/\sqrt{3} \approx 0.845$.

3 Proof

Let X, Y be independent and uniform on $B = B_1 \subset \mathbb{R}^n$, $n \geq 1$.

Lemma 2 (radial moments). $\mathbb{E}|X|^k = n/(n+k)$ for $k \in \mathbb{N}$. Hence $\text{Var}|X| = \frac{n}{n+2} - \frac{n^2}{(n+1)^2} = \frac{n}{(n+1)^2(n+2)} > 0$.

Proof. Polar coordinates give $\int_{\mathbb{R}^n} f(|x|) dx = n|B| \int_0^\infty y^{n-1} f(y) dy$, where $|B|$ is the volume of B . This is Mathlib's `integral_fun_norm_addHaar`. Take $f(y) = y^k \mathbf{1}_{y < 1}$ and divide by $|B|$. This gives $\mathbb{E}|X|^k = n \int_0^1 y^{n-1+k} dy = n/(n+k)$. $\square$

Lemma 3 (pointwise inequality). If $|x|, |y| \leq 1$ and $c \in \mathbb{R}$, then $\langle x, y \rangle^2 \leq 2((|x - y| - c)^2 + (|x + y| - c)^2)$.

Proof. Put $a = |x - y|$ and $b = |x + y|$. Then $b^2 - a^2 = 4\langle x, y \rangle$ and $0 \leq a, b \leq 2$. Therefore $16\langle x, y \rangle^2 = (b - a)^2(b + a)^2 \leq 16(b - a)^2$. Finally $2((a - c)^2 + (b - c)^2) - (b - a)^2 = (a + b - 2c)^2 \geq 0$. $\square$

Lemma 4. $\mathbb{E}\langle X, Y \rangle^2 \leq 4 \text{Var}|X - Y|$.

Proof. Let $c = \mathbb{E}|X - Y|$. The uniform law on B is invariant under $y \mapsto -y$, so (X, Y) and $(X, -Y)$ have the same joint law. Hence $\mathbb{E}(|X + Y| - c)^2 = \mathbb{E}(|X - Y| - c)^2 = \text{Var}|X - Y|$. Integrate Lemma 3, which applies almost surely because $|X|, |Y| < 1$. $\square$

Lemma 5. $\mathbb{E}\langle X, Y \rangle^2 \geq (\mathbb{E}|X|^2)^2/n = n/(n+2)^2$.

Proof. Let $m_{ij} = \mathbb{E}X_i X_j$. Expanding and using independence, $\mathbb{E}\langle X, Y \rangle^2 = \sum_{i,j} \mathbb{E}[X_i X_j] \mathbb{E}[Y_i Y_j] = \sum_{i,j} m_{ij}^2 \geq \sum_i m_{ii}^2$. By Cauchy-Schwarz, $\sum_i m_{ii}^2 \geq (\sum_i m_{ii})^2/n = (\mathbb{E}|X|^2)^2/n$. Then use Lemma 2 with $k = 2$. $\square$

Proof of Theorem 1. Lemmas 4 and 5 give $\text{Var}|X - Y| \geq n/(4(n+2)^2)$. Dividing by $\text{Var}|X| = n/((n+1)^2(n+2))$ gives $a_n \geq (n+1)^2/(4(n+2))$. Moreover $8(n+1)^2 \geq 4n(n+2)$, so $a_n \geq n/8 \rightarrow \infty$. Suppose $a_n - L = O(1/n)$ with $L = 2 - 2/\sqrt{3}$. Since $1/n \rightarrow 0$, this would force $a_n \rightarrow L$, a contradiction.

For radius $r > 0$, the uniform law on B_r is the push-forward of the uniform law on B_1 under $x \mapsto rx$. This follows from $\text{vol}(rA) = r^n \text{vol}(A)$. So $X' = X/r$ and $Y' = Y/r$ are independent and uniform on B_1 , with $\text{Var}|X - Y| = r^2 \text{Var}|X' - Y'|$ and $\text{Var}|X| = r^2 \text{Var}|X'|$. The ratio is therefore unchanged. $\square$

4 Lean formalisation

File `lean/Conjecture7793/Basic.lean`, namespace `C7793`, built with Lean v4.33.1 and Mathlib. Its only import is Mathlib.

- `ballUnif n := ProbabilityTheory.cond volume (ball (0 : EuclideanSpace R (Fin n)) 1)` and `ballUnifR n r` (radius r): the uniform probability measures.
- `ratio n := variance (fun p => norm (p.1 - p.2)) ((ballUnif n).prod (ballUnif n)) / variance (fun x => norm x) (ballUnif n)`.
- `moment:  $\mathbb{E}|X|^k = n/(n+k)$ . var_norm:  $\text{Var}|X| = n/((n+1)^2(n+2))$ .`

- `pointwise, inner_sq_le_var, inner_sq_ge, var_dist_ge`: Lemmas 3–5, and $\text{Var } |X - Y| \geq n/(4(n+2)^2)$.
- `ratio_ge (n) (hn : 1 <= n) : (n+1)^2/(4(n+2)) <= ratio n`.
- `ratio_tendsto_atTop : Tendsto ratio atTop atTop; no_finite_limit (L)`.
- `not_conjecture : not ((fun n => ratio n - (2 - 2/sqrt 3)) = 0[atTop] (fun n => 1/n))`.
- `not_conjecture_rv`: the same negation for arbitrary probability spaces (Ω_n, P_n) , measurable X_n, Y_n with `IndepFun (X n) (Y n) (P n)`, laws $(P n).map (X n) = \text{ballUnifR } n (r n)$ and the same for Y_n , and any radii $r_n > 0$. It uses `map_smul_ballUnif` (scaling of the uniform law) and `ratio_eq_of_indep`.

Axiom audit. `#print axioms` for `not_conjecture, not_conjecture_rv, ratio_ge, ratio_tendsto_atTop` and `no_finite_limit` reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.